

Assignment-1

Title: Average Total Sales All Branch and All Year

➤ Step 1: Loading the Dataset

- First, Tableau Desktop Public Edition is opened and the supermarket_sales dataset is loaded. This dataset contains sales-related information such as product line, branch, date, and total sales amount.

➤ Purpose:

- To make the data available in Tableau for analysis and visualization.

➤ Step 2: Creating a New Worksheet

- After loading the dataset, a new worksheet (Sheet 1) is selected to create the visualization.

➤ Purpose:

- To build and organize the chart in a separate working area.

➤ Step 3: Selecting Product Line as Dimension

- The Product line field is dragged to the Columns shelf.
- This displays different product categories along the horizontal axis.

➤ Purpose:

- To compare total sales across various product lines.

➤ Step 4: Selecting Total Sales as Measure

- The Total field is dragged to the Rows shelf.
- Tableau automatically aggregates it as SUM(Total), which represents the total sales for each product line.

➤ Purpose:

- To calculate and display total revenue generated by each product category.

➤ Step 5: Choosing Bar Chart

- From the Marks card, a bar chart is selected.

➤ Purpose:

- A bar chart makes it easy to visually compare sales values between product lines.

➤ Step 6: Applying Branch Filter

- The Branch field is added to the Filters shelf.
- All branches (A, B, and C) are selected, and the filter is displayed.

➤ **Purpose:**

- To allow users to analyze sales data for specific branches or all branches together.

➤ **Step 7: Applying Year Filter**

- The Date field is added to the Filters shelf and filtered by Year.
- The year 2019 is selected.

➤ **Purpose:**

- To restrict the analysis to a specific time period.

➤ **Step 8: Enabling Data Labels**

- Data labels are enabled from the Marks → Label option, showing the sales value on top of each bar.

➤ **Purpose:**

- To clearly display exact numerical values for better understanding.

➤ **Step 9: Sorting the Data**

- The bars are sorted in descending order based on total sales.

➤ **Purpose:**

- To highlight the highest and lowest performing product lines.

➤ **Step 10: Formatting the Visualization**

- A suitable color is applied to the bars, the background is lightly shaded, and gridlines are removed.

➤ **Purpose:**

- To improve readability and give the chart a clean, professional appearance.

➤ **Step 11: Adding Title**

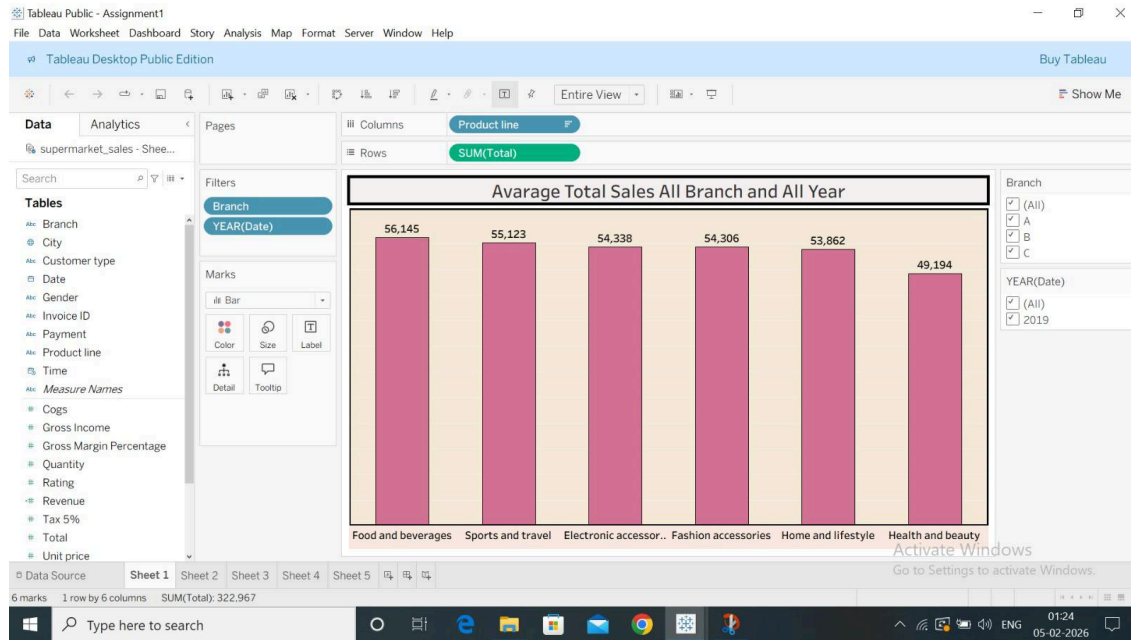
- A descriptive title is added:
- “Average Total Sales All Branch and All Year”

➤ **Purpose:**

- To clearly explain what the visualization represents.

➤ **Step 12: Final Output and Interpretation**

- The final bar chart shows total sales for each product line across all branches in the selected year.
- It helps identify which product categories generate the highest and lowest revenue.



Title: Product Line Wise Sales

➤ Step 1: Load the Dataset

- Open Tableau Desktop Public Edition and load the supermarket_sales dataset.
- The dataset contains sales information such as product line, total sales, branch, and date.

➤ Purpose:

- To make the sales data available for visualization.

➤ Step 2: Create a New Worksheet

- After loading the dataset, select a new worksheet (for example, Sheet 2).
- This worksheet is used to create the pie chart.

➤ Purpose:

- To organize and build a separate visualization.

➤ Step 3: Select Chart Type

- From the Marks card, change the mark type to Pie.

➤ Purpose:

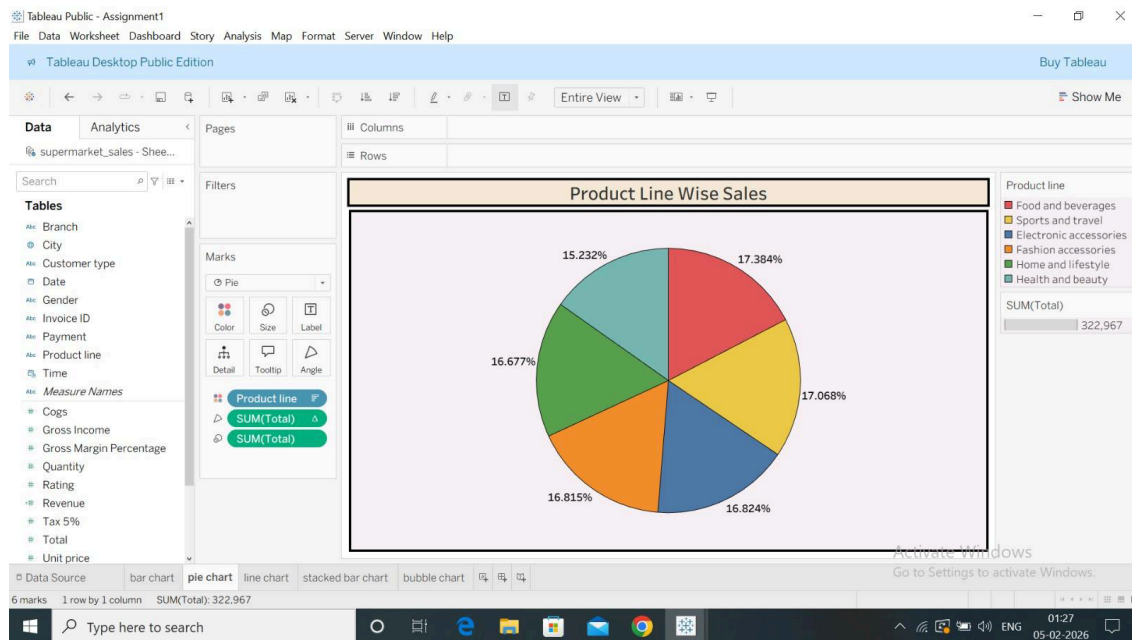
- A pie chart is used to show the proportional contribution of each product line to total sales.

➤ Step 4: Add Product Line Dimension

- Drag Product line from the Data pane and drop it into the Color option in the Marks card.
- **Purpose:**
 - Each product line is displayed as a separate slice with a different color.
- **Step 5: Add Total Sales Measure**
 - Drag Total from Measures and drop it into:
 - Angle
 - Size
 - Ensure the aggregation is SUM(Total).
- **Purpose:**
 - The size and angle of each slice represent the total sales of that product line.
- **Step 6: Show Percentage Labels**
 - Drag SUM(Total) again to the Label option in the Marks card.
 - Click Label
 - Check Show Mark Labels
 - Change label format to Percentage
- **Purpose:**
 - To clearly display each product line's percentage contribution to total sale.
- **Step 7: Adjust Pie Size**
 - Use the Size slider in the Marks card to increase the pie chart size.
- **Purpose:**
 - To improve visibility and readability of the chart.
- **Step 8: Formatting the Chart**
 - Apply different colors to each product line automatically
 - Set a light background color
 - Remove unnecessary gridlines
 - Add borders for a clean look
- **Purpose:**
 - To make the visualization visually appealing and professional.
- **Step 9: Add Title**
 - Add a title to the chart: "Product Line Wise Sales"
- **Purpose:**
 - To clearly describe what the visualization represents.

➤ Step 10: Final Interpretation

- The pie chart shows the percentage share of total sales contributed by each product line:
 - Food and Beverages
 - Sports and Travel
 - Electronic Accessories
 - Fashion Accessories
 - Home and Lifestyle
 - Health and Beauty
- This helps in identifying which product line contributes the most and least to overall sales.



Title: Month Wise Total Sales (All Year)

➤ Step 1: Load the Dataset

- Open Tableau Desktop Public Edition and load the supermarket_sales dataset.
- This dataset contains sales details such as date, product line, branch, and total sales.

➤ Purpose:

- To bring the sales data into Tableau for analysis.

➤ Step 2: Create a New Worksheet

- After loading the dataset, open a new worksheet (for example, Sheet 3).
- This worksheet is used to create the line chart.

➤ Purpose:

- To build a separate visualization for monthly sales analysis.

➤ Step 3: Select Date at Month Level

- Drag the Date field to the Columns shelf.
- Change the date level to MONTH(Date) so that sales are displayed month-wise.

➤ Purpose:

- To analyze how total sales change from month to month.

➤ Step 4: Add Total Sales Measure

- Drag Total from the Measures pane to the Rows shelf.
- Ensure the aggregation is set to SUM(Total).

➤ Purpose:

- To calculate the total sales amount for each month.

➤ Step 5: Choose Line Chart

- From the Marks card, select Line as the chart type.

➤ Purpose:

- A line chart is ideal for showing trends and changes over time.

➤ Step 6: Apply Year Filter

- Drag Date again to the Filters shelf.
- Select Years and choose 2019.
- Show the filter if required.

➤ Purpose:

- To restrict the analysis to sales data from the year 2019 only.

➤ **Step 7: Enable Data Labels**

- In the Marks card, click Label and enable Show Mark Labels.
- This displays the total sales value for each month.

➤ **Purpose:**

- To clearly show exact monthly sales values on the chart.

➤ **Step 8: Formatting the Line Chart**

- Adjust line thickness for better visibility
- Apply color gradient based on SUM(Total)
- Set a light background color
- Remove unnecessary gridlines

➤ **Purpose:**

- To improve readability and give the chart a clean, professional look.

➤ **Step 9: Add Chart Title**

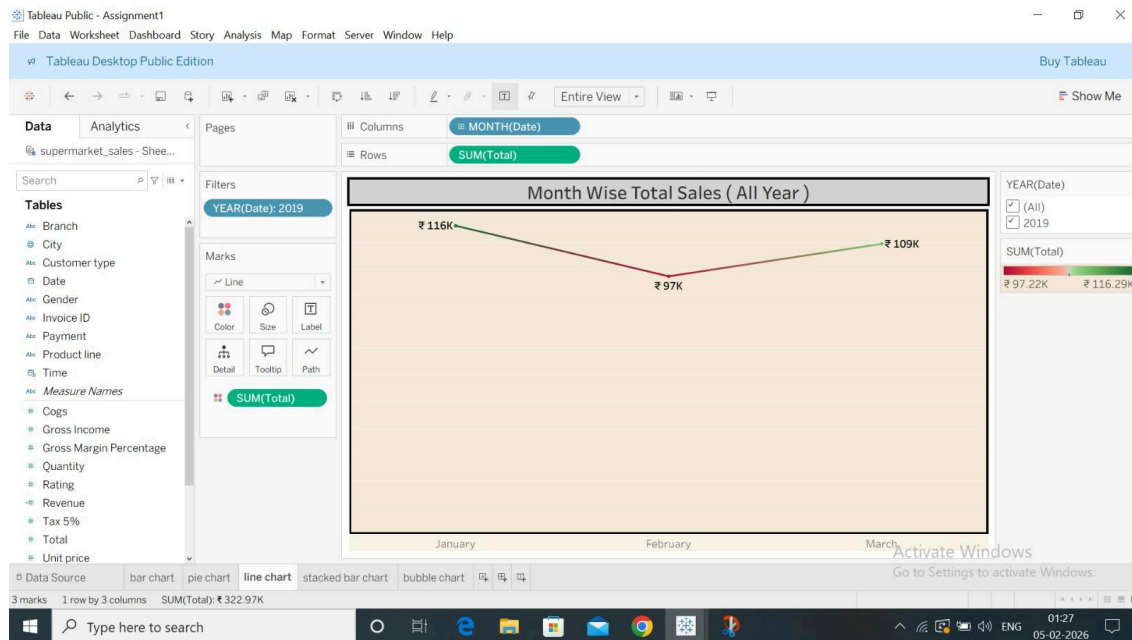
- Add the title: “Month Wise Total Sales (All Year)”

➤ **Purpose:**

- To clearly describe the content of the visualization.

➤ **Step 10: Final Interpretation**

- The line chart shows how total sales vary across months:
- January shows higher sales
- February shows a decrease
- March shows an increase again
- This indicates sales fluctuations over time.



Title: Total Sales Across Product Line by City

➤ Step 1: Load the Dataset

- Open Tableau Desktop Public Edition and load the supermarket_sales dataset.
- The dataset contains information such as Product line, City, Branch, Date, and Total sales.

➤ Purpose:

- To make the data available for analysis and visualization.

➤ Step 2: Create a New Worksheet

- After loading the dataset, open a new worksheet (for example, Sheet 4).
- This worksheet is used to create the stacked bar chart.

➤ Purpose:

- To organize the visualization separately.

➤ Step 3: Add Product Line to Rows

- Drag the Product line field to the Rows shelf.

- This displays different product categories on the vertical axis.
- **Purpose:**
 - To compare sales for each product line.
- **Step 4: Add Total Sales to Columns**
 - Drag the Total field to the Columns shelf.
 - Ensure it is aggregated as SUM(Total).
- **Purpose:**
 - To calculate and display the total sales for each product line.
- **Step 5: Add City for Stacking**
 - Drag the City field to the Color option in the Marks card.
 - This stacks each bar by city (Yangon, Naypyitaw, Mandalay).
- **Purpose:**
 - To show how each city contributes to the total sales of a product line.
- **Step 6: Apply Branch Filter**
 - Drag the Branch field to the Filters shelf.
 - Select all branches (A, B, and C) and show the filter if required.
- **Purpose:**
 - To include sales data from all branches.
- **Step 7: Apply Year Filter**
 - Drag the Date field to the Filters shelf.
 - Select Year and choose 2019.
- **Purpose:**
 - To restrict the analysis to a specific year.
- **Step 8: Enable Data Labels**
 - In the Marks card, click Label and enable Show Mark Labels.
 - This displays sales values for each city segment.
- **Purpose:**
 - To clearly show exact sales figures.
- **Step 9: Formatting the Chart**
 - Apply different colors for each city
 - Remove unnecessary gridlines
 - Adjust bar size and background color
 - Ensure readability of labels

➤ **Purpose:**

- To make the visualization clear and professional.

➤ **Step 10: Add Chart Title**

- Add the title: “Total Sales Across Product Line by City”

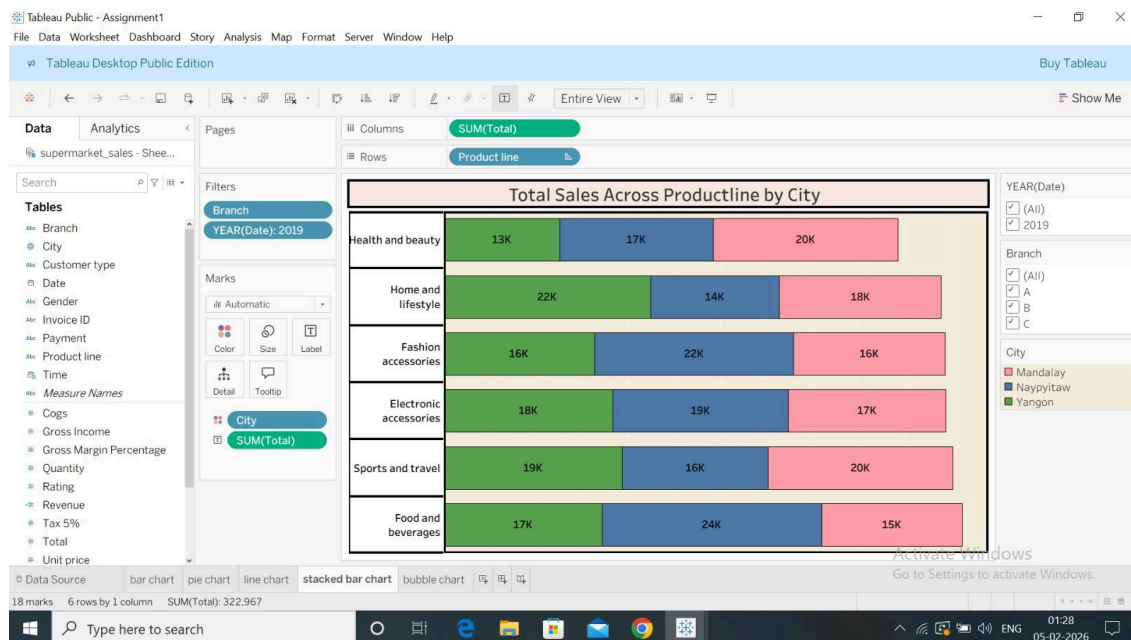
➤ **Purpose:**

- To clearly describe what the visualization represents.

➤ **Step 11: Final Interpretation**

The stacked bar chart shows:

- Total sales of each product line
- City-wise contribution within each product line
- Comparison of sales performance across cities and product categories



Title: Revenue by Product Line

➤ **Step 1: Load the Dataset**

- Open Tableau Desktop Public Edition and load the supermarket_sales dataset.

- This dataset includes fields such as Product line, City, Revenue, Date, and Branch.
- **Purpose:**
 - To make the sales and revenue data available for visualization.
- **Step 2: Create a New Worksheet**
 - After loading the dataset, open a new worksheet (for example, Sheet 5).
 - This worksheet is used to create the bubble chart.
- **Purpose:**
 - To build a separate visualization for revenue analysis.
- **Step 3: Select Chart Type**
 - From the Marks card, change the mark type to Circle.
- **Purpose:**
 - A circle (bubble) chart helps compare values using bubble size.
- **Step 4: Add Revenue Measure**
 - Drag Revenue from the Measures pane and drop it into:
 - Size, Label
 - Ensure the aggregation is SUM(Revenue).
- **Purpose:**
 - The bubble size represents the total revenue generated.
- **Step 5: Add Product Line**
 - Drag Product line to the Label option in the Marks card.
- **Purpose:**
 - To display the product line name inside each bubble.
- **Step 6: Add City for Color**
 - Drag City to the Color option in the Marks card.
 - Each city (Yangon, Naypyitaw, Mandalay) is shown using a different color.
- **Purpose:**
 - To differentiate revenue contributions by city.
- **Step 7: Adjust Bubble Size**
 - Use the Size slider in the Marks card to increase or decrease bubble size.
- **Purpose:**
 - To improve visibility and avoid overlap.
- **Step 8: Formatting the Chart**

- Apply distinct colors for each city
- Set a light background color
- Adjust font size for labels
- Ensure bubbles are clearly visible

➤ **Purpose:**

- To make the chart visually appealing and easy to understand.

➤ **Step 9: Add Chart Title**

Add the title: “Revenue by Product Line”

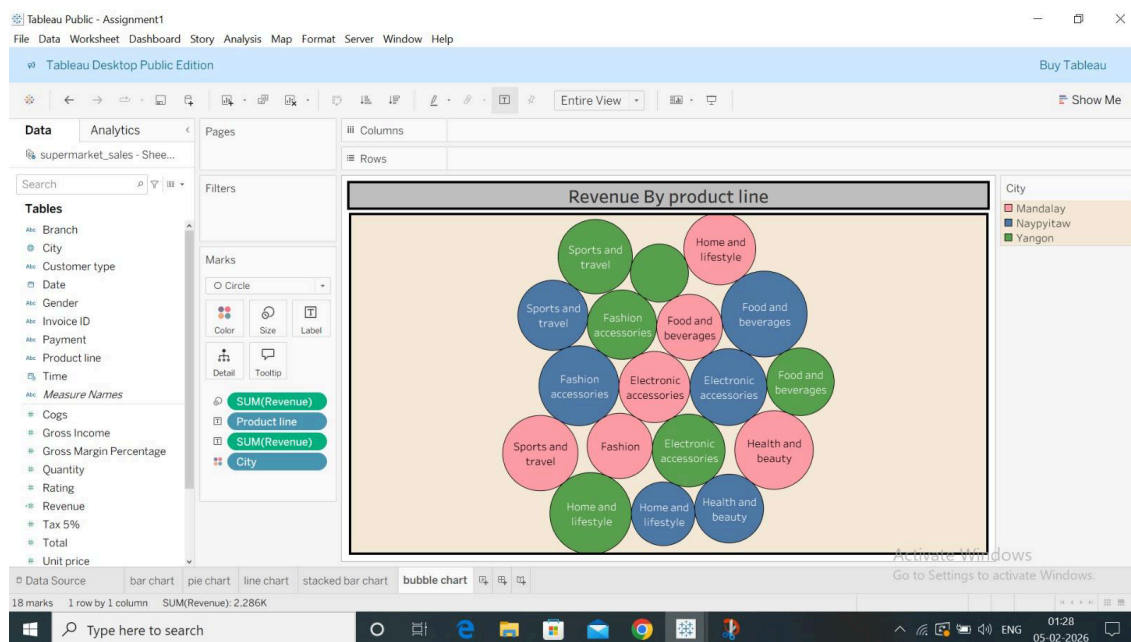
➤ **Purpose:** Total Sales Across Product Line by City

- To clearly describe what the visualization represents.

➤ **Step 10: Final Interpretation**

The bubble chart shows:

- Revenue generated by each product line
- Comparison of revenue across different cities
- Larger bubbles indicate higher revenue
- This helps quickly identify high-revenue and low-revenue product lines.



Assignment 2:

Area chart

- **Step 1: Load the Dataset**

Open Tableau Desktop Public Edition and load the supermarket sales dataset.

- **Purpose:**

To access sales-related data such as Date, Product line, City, and Total sales.

- **Step 2: Create a New Worksheet**

Open a new worksheet (for example, Sheet 5) to build the visualization.

- **Purpose:**

To design a separate chart for monthly sales analysis.

- **Step 3: Add Date to Columns**

Drag the Date field to the Columns shelf.

Change the date format to MONTH(Date).

- **Purpose:**

To display sales data month-wise (January, February, March).

- **Step 4: Add Total Sales to Rows**

Drag the Total measure to the Rows shelf.

Ensure the aggregation is SUM(Total).

- **Purpose:**

To calculate total sales for each month.

- **Step 5: Change Mark Type to Area**

From the Marks card, select Area as the chart type.

- **Purpose:**

An area chart shows trends and cumulative values clearly over time.

- **Step 6: Add Product Line to Color**

Drag Product line to the Color option in the Marks card.

- **Purpose:**

To divide total sales by different product lines such as:

Electronic accessories

Fashion accessories

Food and beverages

Health and beauty

Home and lifestyle

Sports and travel

Each product line is displayed in a different color.

- **Step 7: Apply Filters**

Drag City to the Filters shelf

Drag YEAR(Date) to the Filters shelf and select 2019

- **Purpose:**

To analyze sales data for a specific year and city if required.

- **Step 8: Enable Labels**

Turn on Label from the Marks card to display sales values (in K format).

- **Purpose:**

To show exact sales values for better understanding.

- **Step 9: Format the Chart**

Adjust colors for clarity

Apply background shading

Improve font size and alignment

Ensure the stacked areas are clearly visible

- **Purpose:**

To make the visualization neat and professional.

- **Step 10: Add Chart Title**

Give the title: “Month Wise Total Sales Across Product Line”

- **Purpose:**

To clearly explain what the chart represents.

Final Interpretation

The chart shows total sales for each month

Sales are broken down by product line

The stacked area indicates cumulative monthly sales

Helps identify high-performing and low-performing product lines across months

Word cloud

- **Step 1: Load the Dataset**

Open Tableau Desktop Public Edition and load the supermarket sales dataset.

- **Purpose:**

To access sales information such as Product line, Branch, and Total sales.

Step 2: Create a New Worksheet

Open a new worksheet to design the word cloud visualization.

- **Purpose:**

To analyze total sales by product line in a visual text format.

- **Step 3: Change Mark Type to Text**

From the Marks card, select Text as the mark type.

- **Purpose:**

Text marks are required to create a word cloud.

- **Step 4: Add Product Line**

Drag Product line to the Text option in the Marks card.

- **Purpose:**

To display each product line as a word in the visualization.

- **Step 5: Add Total Sales to Size**

Drag Total to the Size option in the Marks card.

Ensure the aggregation is SUM(Total).

- **Purpose:**

The size of each word represents the total sales value.

Larger words indicate higher sales.

- **Step 6: Add Branch to Color**

Drag Branch to the Color option in the Marks card.

Purpose:

To distinguish sales from different branches (A, B, C) using colors.

- **Step 7: Adjust Word Size and Layout**

Use the Size slider to adjust word sizes and ensure all words are visible without overlap.

- **Purpose:**

To improve readability of the word cloud.

- **Step 8: Format the Visualization**

Apply suitable colors for each branch

Adjust font size and background

Ensure the text is clearly visible

- **Purpose:**

To make the word cloud visually attractive and professional.

- **Step 9: Add Chart Title**

Add the title: “Total Sales by Product Line”

- **Purpose:**

To clearly explain what the visualization shows.

Final Interpretation

Each word represents a product line

Larger words indicate higher total sales

Different colors show sales from different branches

Helps quickly identify top-selling and low-selling product lines

Text Table

- **Step 1: Load the Dataset**

Open Tableau Desktop Public Edition and load the Supermarket Sales dataset.

Purpose:

To access fields such as Product line, City, Date, and Revenue required for analysis.

- **Step 2: Create a New Worksheet**

Create a new worksheet to design the text table (also called a crosstab).

- **Purpose:**

To display detailed numerical values in a tabular format.

Step 3: Add Date to Columns

Drag the Date field to the Columns shelf.

Change it to MONTH(Date).

- **Purpose:**

To display revenue values month-wise (January, February, March).

- **Step 4: Add Product Line and City to Rows**

Drag Product line to the Rows shelf

Drag City below Product line in the Rows shelf

- **Purpose:**

To organize the table hierarchically:

by Product line

Then by City within each product line

- **Step 5: Add Revenue Measure**

Create calculation field

Fill the name Revenue

Enter code:[Unit price] * [Quantity]

Click on apply

Drag Revenue to the Text option in the Marks card.

Ensure aggregation is SUM(Revenue).

- **Purpose:**

To display total revenue values for each Product line–City–Month combination.

- **Step 6: Enable Grand Totals**

Go to Analysis → Totals → Show Column Grand Totals

Go to Analysis → Totals → Show Row Grand Totals

- **Purpose:**

Column grand totals show total revenue per month

Row grand totals show total revenue per product line and city

Final grand total shows overall revenue

- **Step 7: Format the Table**

Adjust row and column shading

Improve font size and alignment

Add grid lines for clarity

Format numbers in K (thousands)

- **Purpose:**

To improve readability and make the table professional.

- **Step 8: Sort Data**

Sort Product lines or Cities based on SUM(Revenue) if required.

- **Purpose:**

To easily identify high-performing and low-performing categories.

- **Step 9: Add Worksheet Title**

Add the title: “Month-wise Revenue by Product Line and City”

- **Purpose:**

To clearly explain what the table represents.

Step 10: Final Interpretation

Each row shows a product line and city combination

Each column shows monthly revenue

Grand totals summarize revenue across months and categories

Helps compare:

Revenue across months

Revenue across cities

Revenue across product lines

Funnel chart

- **Step 1: Load the Dataset**

Open Tableau Desktop Public Edition and connect to the Supermarket Sales dataset.

- **Purpose:**

To access sales-related fields such as City and Total Sales.

- **Step 2: Create a New Worksheet**

Click on New Worksheet to create a fresh visualization.

- **Purpose:**

To design a separate chart for city-wise total sales.

- **Step 3: Drag Total Sales to Rows**

Drag the Total measure to the Rows shelf.

Ensure it is aggregated as SUM(Total).

- **Purpose:**

To calculate total sales value for each city.

- **Step 4: Drag City to Marks Card**

Drag the City dimension to the Marks card.

- **Purpose:**

To split total sales based on cities (Naypyitaw, Yangon, Mandalay).

- **Step 5: Change Chart Type to Funnel**

Click on Show Me → select Funnel Chart

(or manually adjust the Marks type to create a funnel-style visualization).

- **Purpose:**

To visually compare sales values in descending order.

- **Step 6: Add Color Encoding**

Drag City to the Color option in the Marks card.

- **Purpose:**

Each city gets a unique color

Makes the funnel visually clear and easy to differentiate

- **Step 7: Add Labels**

Drag SUM(Total) and City to the Label option.

- **Purpose:**

To display: City name

Exact total sales value on each funnel section

- **Step 8: Sort Cities by Total Sales**

Sort City in descending order based on SUM(Total).

Purpose:

To arrange the funnel from highest sales to lowest sales.

- **Step 9: Format the Chart**

Adjust bar size for funnel shape

Improve font size and alignment

Add borders and shading

Format numbers with commas

- **Purpose:**

To make the chart neat, readable, and professional.

Step 10: Add Chart Title

Set the title as:

“Total Sales by City”

- **Purpose:**

To clearly describe what the visualization represents.

Interpretation of the Chart

Naypyitaw has the highest total sales

Yangon is the second highest

Waterfall chart

- **Step 1: Connect to the Dataset**

Open Tableau Desktop Public Edition and connect to the Supermarket Sales dataset.

- **Purpose:**

To access product-wise sales data required for analysis.

- **Step 2: Create a New Worksheet**

Click on New Worksheet to start building the visualization.

- **Purpose:**

To design a separate chart for product-line-wise sales.

- **Step 3: Drag Product Line to Columns**

Drag the Product line dimension to the Columns shelf.

- **Purpose:**

Each product line (Electronic accessories, Fashion accessories, etc.) will appear on the X-axis.

- **Step 4: Drag Total to Rows**

Drag the Total measure to the Rows shelf.

Ensure it is aggregated as SUM(Total).

- **Purpose:**

To calculate the total sales value for each product line.

- **Step 5: Change the Chart Type to Gantt Bar**

In the Marks card, change the mark type to Gantt Bar.

- **Purpose:**

Gantt bars are required to construct a waterfall chart structure.

- **Step 6: Create Running Total Calculation**

Right-click on SUM(Total) → select Quick Table Calculation → choose Running Total.

- **Purpose:**

To show cumulative sales changes across product lines step by step.

- **Step 7: Create a Difference Calculation (Waterfall Effect)**

Create a calculated field (or use table calculation) to show incremental contribution between product lines.

- **Purpose:**

This allows each bar to represent how much each product line adds to the total sales.

- **Step 8: Add Color Encoding**

Drag SUM(Total) to the Color shelf.

- **Purpose:**

Darker color = higher sales contribution

Lighter color = lower sales contribution

Improves visual interpretation

- **Step 9: Add Grand Total Bar**

Enable Grand Total from the Analysis → Totals → Show Column Grand Totals option.

- **Purpose:**

To display the final cumulative total of all product lines on the right side.

- **Step 10: Add Labels**

Drag SUM(Total) to the Label shelf.

Purpose:

To display sales values on each step of the waterfall chart.

- **Step 11: Format the Chart**

Adjust bar size for clarity

Rotate product line labels vertically

Apply shading and borders

Format numbers with commas

- **Purpose:**

To make the chart clean, readable, and professional.

- **Step 12: Add Chart Title**

Set the title as:

“Total Sales by Product Line”

- **Purpose:**

Clearly explains what the visualization represents.

Interpretation of the Chart

Each bar shows the incremental contribution of a product line to total sales

Sales increase progressively from left to right

Sports and Travel and Home and Lifestyle show higher contributions

The Grand Total bar represents overall total sales from all product lines combined